MOHAMED KARIM BELKADI

SOFTWARE ENGINEER | FULL-STACK, APPLIED AI & COMPUTER VISION

Email: belkadikarim36@gmail.com | **City:** Rabat, Morocco | **Portfolio:** karimbelkadi.github.io

PROFILE

Computer Science and Networks engineering graduate (MIAGE) who builds practical web and desktop software, data tools, search systems, and computer-vision applications. Comfortable moving between backend engineering and user-facing features; open to full-stack, backend, and applied AI roles.

PROFESSIONAL EXPERIENCE

Freelance Full-Stack Developer | Independent **Sep 2026 - Present**

- Develop Java and Spring Boot REST API features backed by relational databases, with maintainable service and data layers.
- Add authentication, role-based access, and CRUD workflows to support secure application features.

Final-Year Project Intern | Marokko Biz, Kenitra **Mar - Jun 2026**

- Contributed Laravel and React/TypeScript features to FlexTimer, a web and desktop HR platform; connected clock-in/out events to HR/admin notifications and worked across attendance, schedules, requests, and document workflows.
- Built MarocLoi search features across a large Moroccan law collection, combining BM25 and semantic retrieval with a local RAG assistant for questions in French, Arabic, and English.
- Automated employee PDF records, including certificates and payslips, and added QR-verifiable references.

Web Development Intern | One Fancy, Kenitra **2025 | 2 months**

- Developed PHP/MySQL app features for apartment and client records, reservations, transfers, and payment tracking.

Development Intern | ONEE, Rabat **2024 | 1 month**

- Built a Python desktop ETL workflow to extract and clean PDF data, calculate pumping-station estimates from HMT and flow, and export Excel reports with charts.

PERSONAL PROJECTS

Live-video fire detection | Python, computer vision

Detects flames in live video, marks detected regions, and emails a Gmail alert with a snapshot attached.

Hand-gesture desktop controls | Python, MediaPipe, Random Forest

Classifies five static gestures from MediaPipe hand landmarks with a Random Forest model, mapping them to volume, brightness, and close controls; webcam frames are processed locally and not saved.

Game and 3D modeling (in progress) | Godot, GDScript, Blender

Learning Godot while building a fantasy game direction and creating original environment and character models.

TECHNICAL SKILLS

Programming: Python, Java, PHP, C#, C++, JavaScript/TypeScript

Web and backend: Laravel, React, Angular, Spring Boot, Java EE/J2EE, ASP.NET Core, REST APIs, Electron

AI and data: OpenCV, MediaPipe, Random Forest, RAG, BM25, Pandas, PDF extraction and ETL

Data and tools: MySQL, PostgreSQL, SQL Server, Docker, Git, Azure DevOps, Linux

Exploring: Godot/GDScript and Blender 3D modeling

EDUCATION AND LANGUAGES

Engineering Degree in Computer Science and Networks - MIAGE | EMSI, Morocco **2021 - 2026**

Languages: English (C1), French (B2)

Selected Coursera courses: Machine Learning with Python; Software Design and Project Management; Docker, Kubernetes, and OpenShift.